R18

Code No: 151AA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year I Semester Examinations, July - 2021
MATHEMATICS-I
(Common to EEE, CSE, IT, CSIT, ITE, CE(SE), CSE(CS), CSE(DS), CSE(Networks))

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

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1.a) Find the inverse of the matrix $A = \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$ using Gauss-Jordan method.

b) Find whether the following equations are consistent if so solve them
$x-y+2z=9,\ 2x+y+z=6,\ x+2y+z=3$ [7+8]

2. Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ [15]

3. Find the matrix P which transforms $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ into diagonal form. Hence calculate A^4. [15]

4.a) Examine the convergence of the series $\sum \frac{(n+1)(n+2)}{n^2\sqrt{n}}$

b) Test for the convergence of the series $1+\frac{2}{5}x+\frac{6}{9}x^2+\frac{14}{17}x^3+...+\frac{2^n-2}{2^n+1}x^{n-1}+...\ (x>0)$ [7+8]

5.a) Discuss the nature of the series $\sum \frac{(n+1)^n\, x^n}{x^{n+1}}$

b) Verify whether the series $\frac{x}{1+x}-\frac{x^2}{1+x^2}+\frac{x^3}{1+x^3}-\frac{x^4}{1+x^4}+...\infty\ (0<x<1)$ is absolutely convergent or conditionally convergent. [7+8]

6.a) Expand $e^{\sin x}$ by Maclaurin's series

b) Prove that $\beta(m+1,n)+\beta(m,n+1)=\beta(m,n)$. [7+8]

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7.a) If $u = x^2 - 2y, v = x + y + z. w = x - 2y + 3z$ then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$.

b) If $f(x,y) = ln(x^2 + y^2) + tan^{-1}\left(\frac{y}{x}\right)$ then prove that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0.$ [7+8]

8.a) A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction.

b) If $u = \frac{x+y}{1-xy}$ and $\theta = \tan^{-1} x + \tan^{-1} y$, then find $\frac{\partial(u,\theta)}{\partial(x,y)}$. Hence prove that u and θ are functionally dependent. Also find the relation between them. [7+8]

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